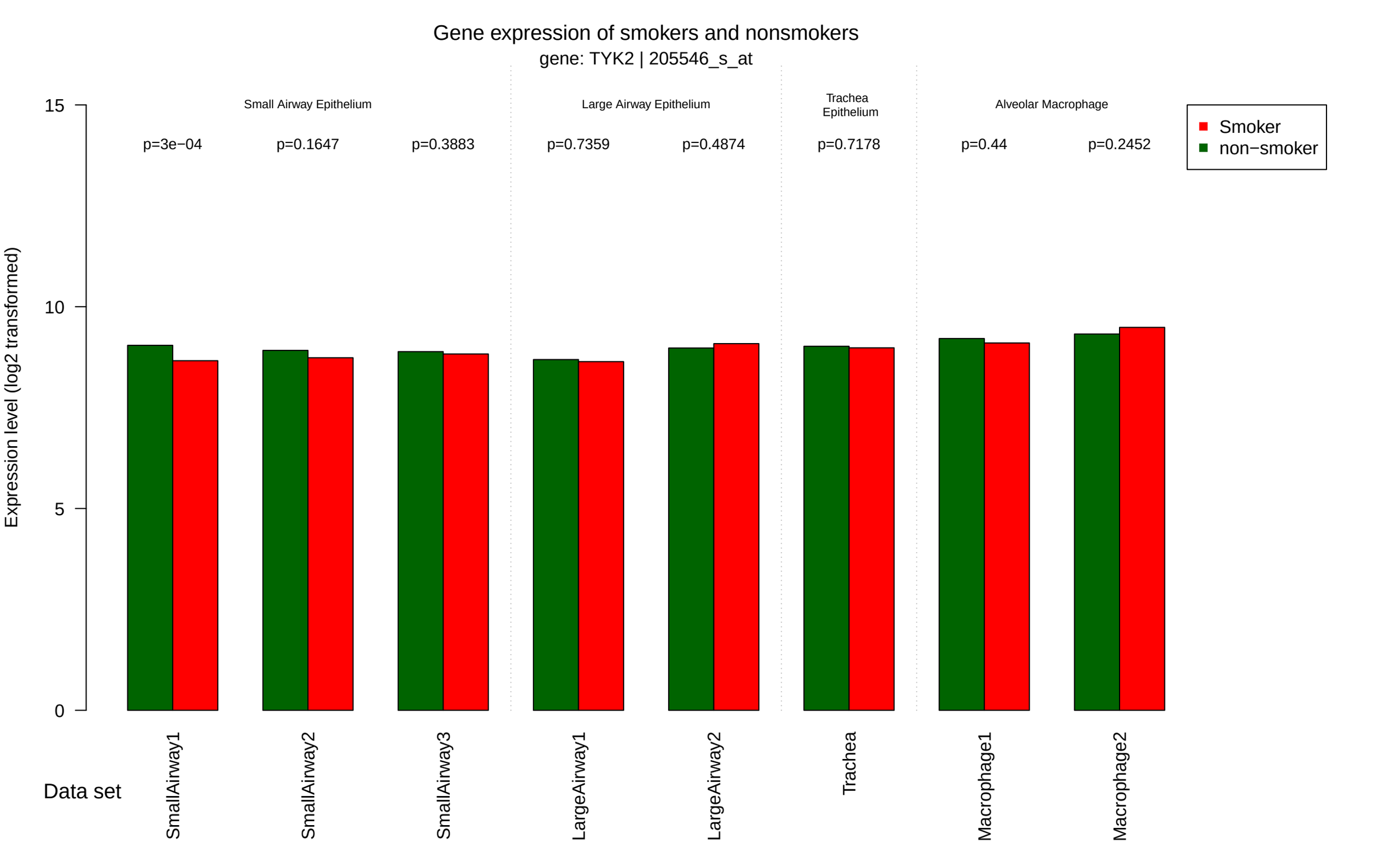
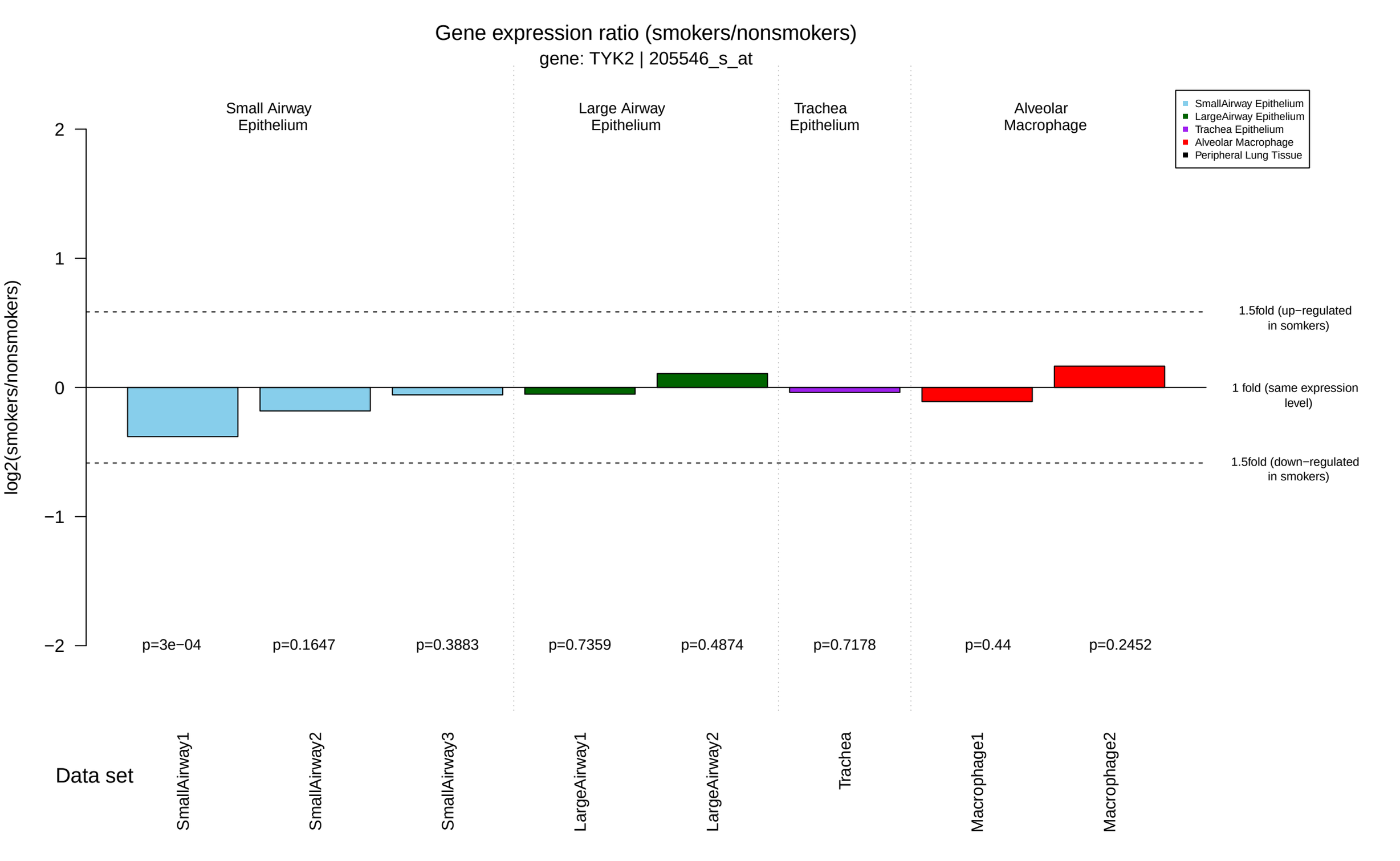
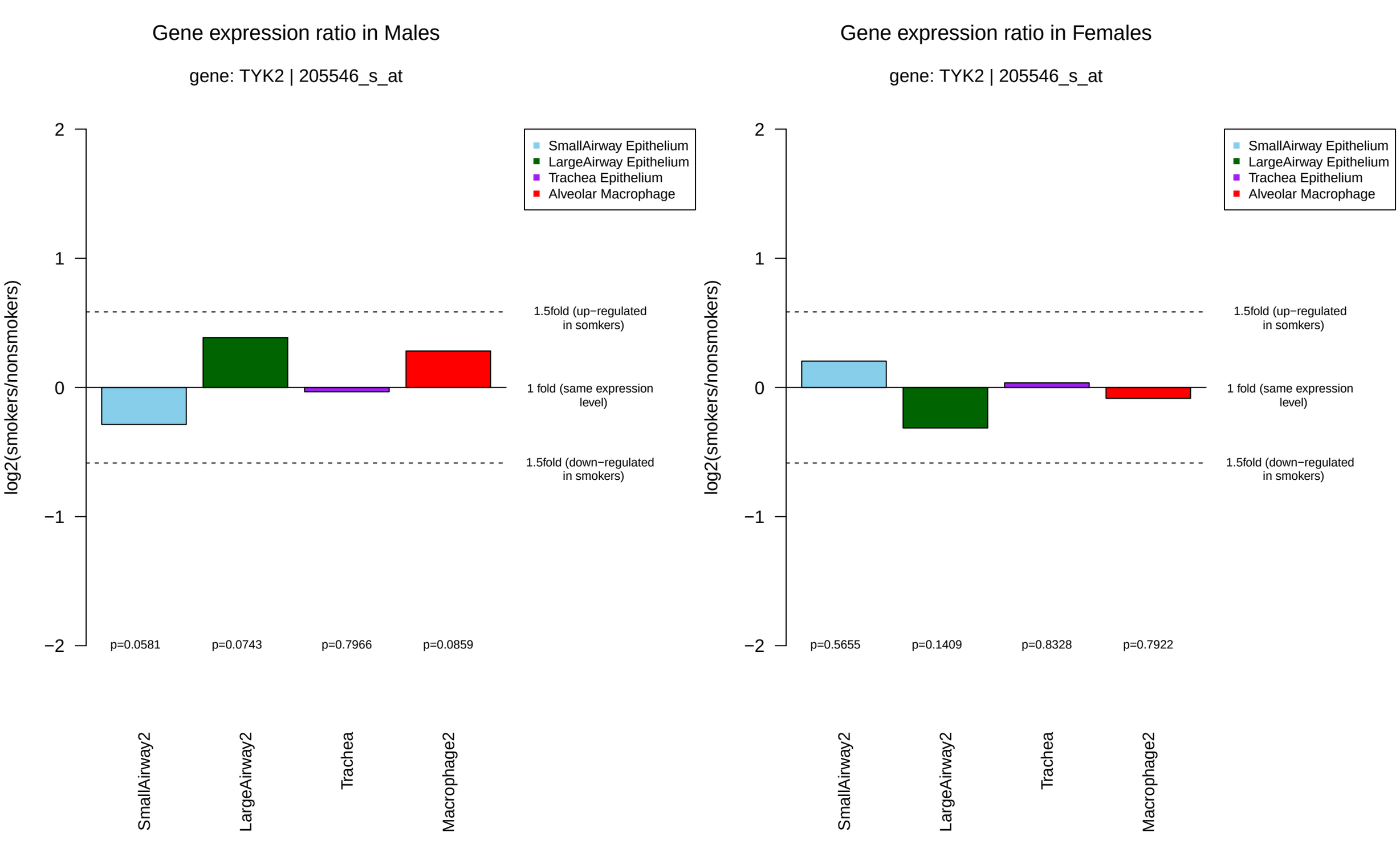
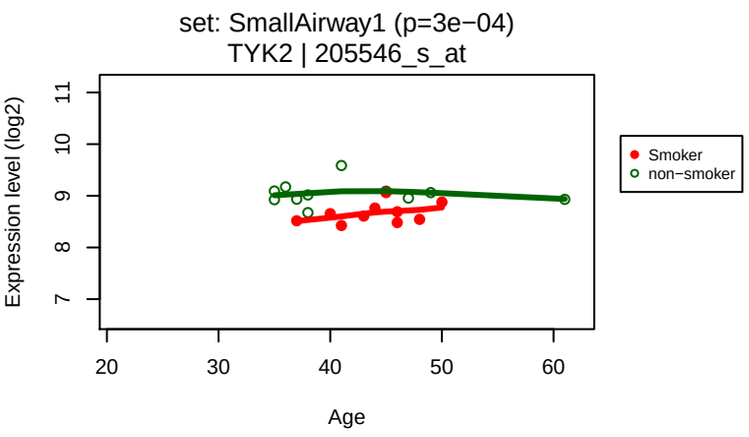
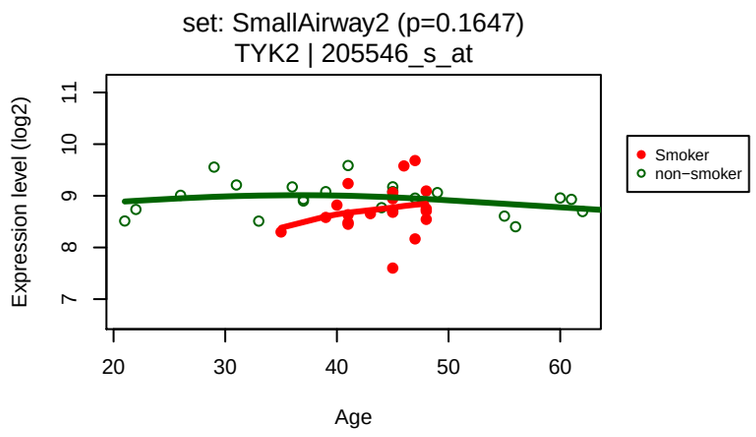
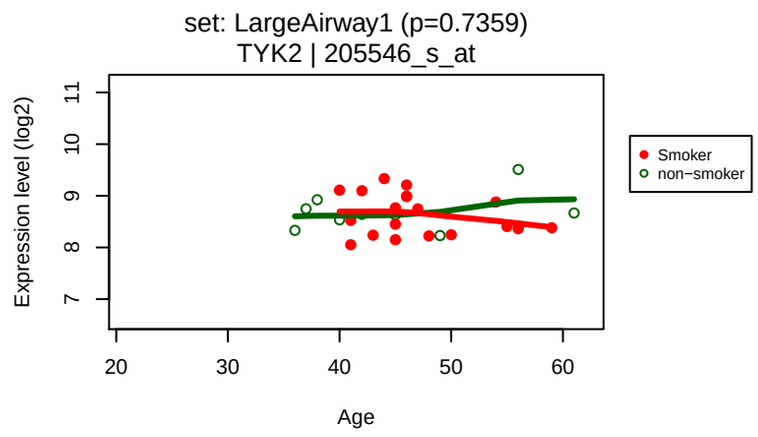

Gene expression of smokers and nonsmokers
gene: TYK2 | 205546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=3e−04
p=0.1647
p=0.3883
p=0.7359
p=0.4874
p=0.7178
p=0.44
p=0.2452
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

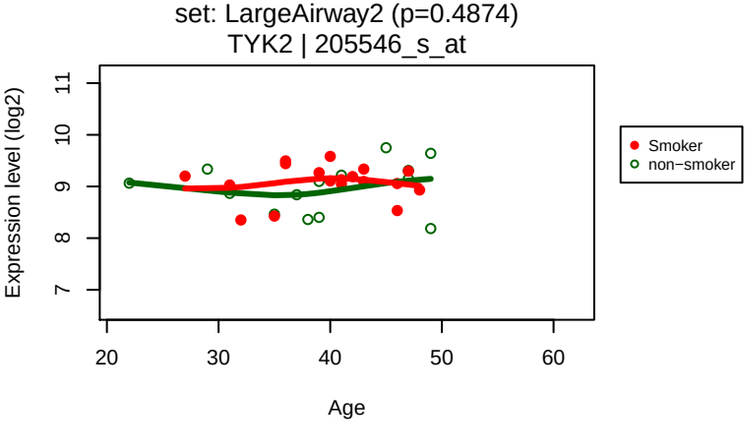

Gene expression ratio (smokers/nonsmokers)
gene: TYK2 | 205546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=3e−04
p=0.1647
p=0.3883
p=0.7359
p=0.4874
p=0.7178
p=0.44
p=0.2452
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

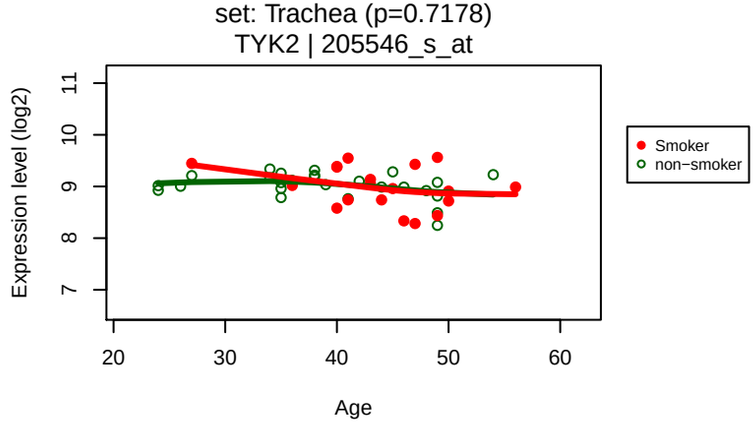

Gene expression ratio in Males
gene: TYK2 | 205546_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0581
p=0.0743
p=0.7966
p=0.0859
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TYK2 | 205546_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5655
p=0.1409
p=0.8328
p=0.7922
SmallAirway2
LargeAirway2
Trachea
Macrophage2

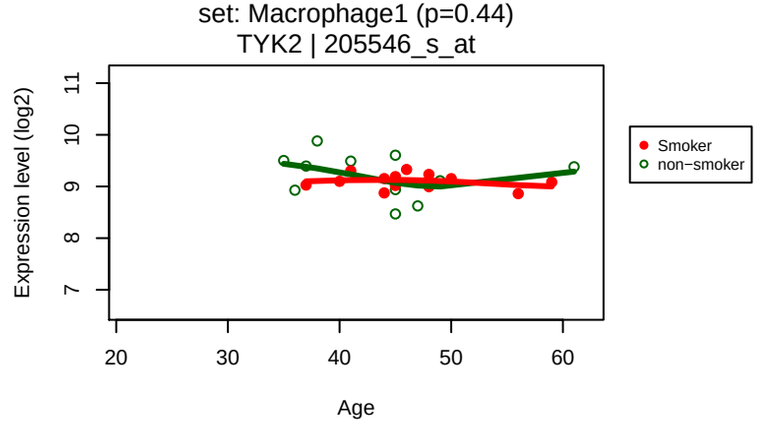

set: SmallAirway1 (p=3e−04)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker

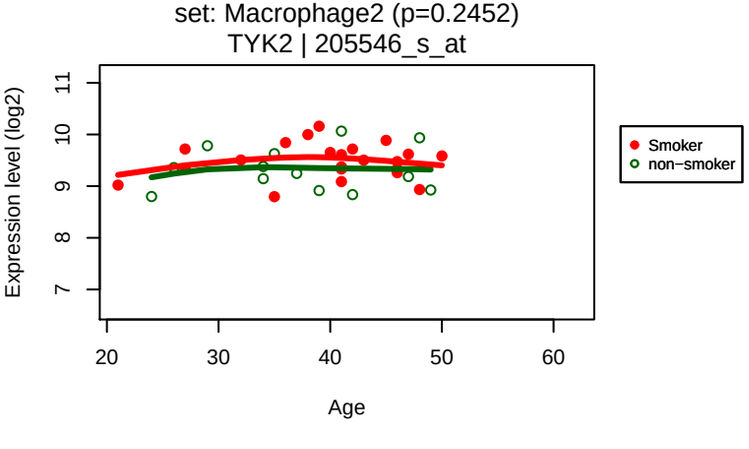

set: SmallAirway2 (p=0.1647)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker

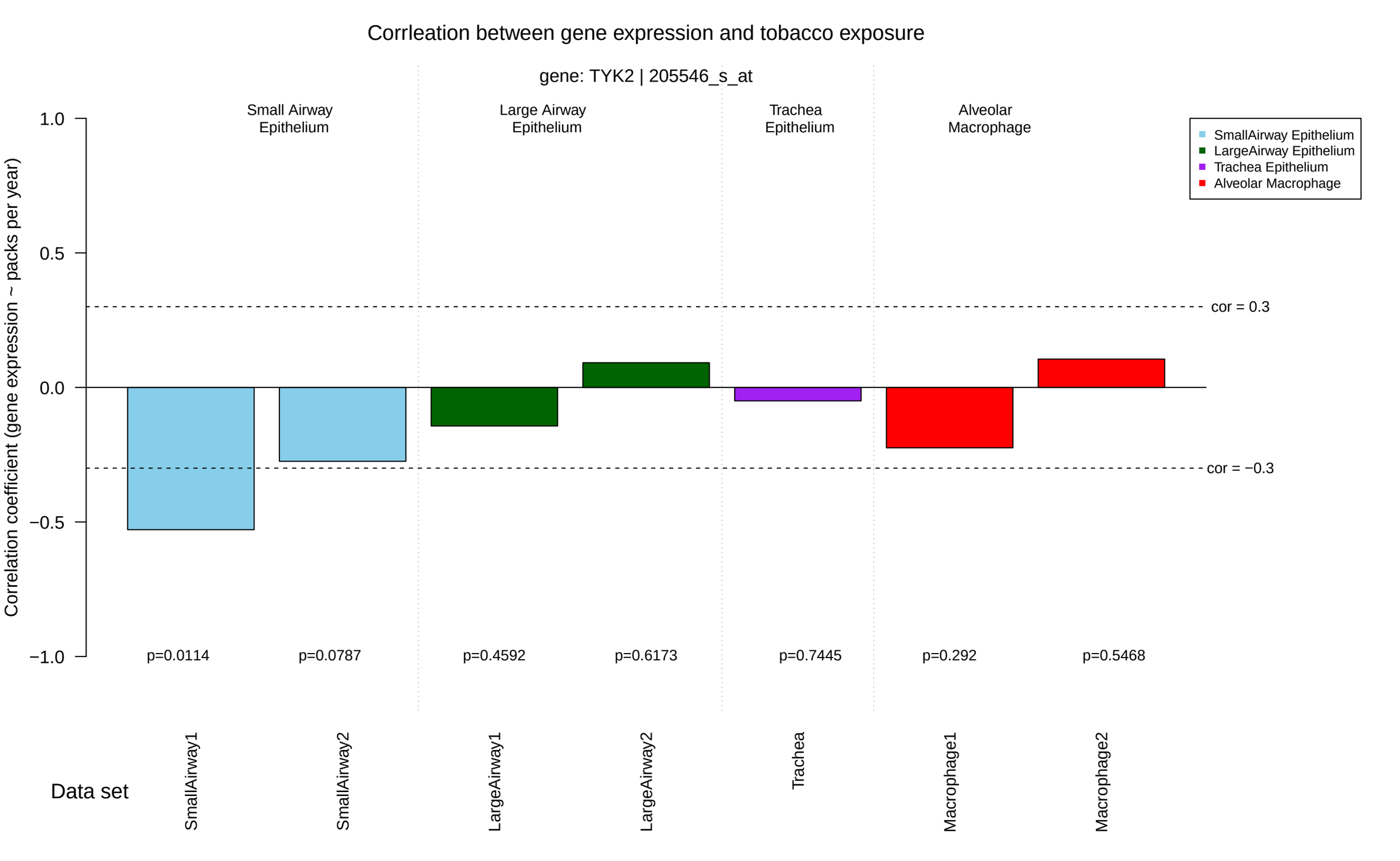

set: LargeAirway1 (p=0.7359)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4874)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7178)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.44)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2452)
TYK2 | 205546_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TYK2 | 205546_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0114
p=0.0787
p=0.4592
p=0.6173
p=0.7445
p=0.292
p=0.5468
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2